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PROJECT REPORT

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Comparative Evaluation of Lightweight YOLO Models for Tree Detection and Counting in UAV Imagery

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Abstract -**Individual tree detection from UAV images is crucial to forest conservation, plantation management and carbon accounting. A YOLOv9t model achieved the lowest error in tree counts (MAE = 2.85 trees/image). The highest detection accuracy (mAP@0.5 = 0.6083) was achieved using a YOLO11n model on the test set. Six models were trained on the publicly available 'Tree-Top-View' dataset containing 1,191 images (116 validation, 55 test) with 443 known trees using local CPU training budgets. This study investigates whether lightweight YOLO object detectors can be used to automatically count trees in a single stage detector by predicting single class bounding boxes. Manual tree counting in large aerial orthomosaics is tedious, labour intensive and prone to inconsistencies caused by touching/overlapping tree crowns. Detection accuracy across all models dropped significantly when detecting on cross-domain images (mAP@0.5 = 0.17). Severe undercounting occurred in heavily clustered, multi-layered canopies. However, detailed error analysis shows that the tree count accuracy of YOLOv9t relies on false positives and negatives cancelling each-other out. Interestingly, the older anchor based YOLOv7-tiny model performed better than the newer anchor-free models (mAP@0.5 = 0.3058). Upgrading the inference resolution from 416 to 640px improved recall by up to 36% without further training. It was shown that the lightweight YOLO models can accurately count trees in open and planted forests however direct zero-shot transfer failed to detect trees in dense native forests. Models were tested for generalisation by testing zero-shot on the external SelvaBox tropical rainforest benchmark containing 100 orthomosaics and 8111 trees.**

Index Terms, Drone remote sensing, individual tree crown detection, automated tree counting, YOLO, domain shift, tropical rainforest, aerial survey, in many respects.

I. INTRODUCTION

Knowing how many trees stand in a forest or plantation is essential to forestry management, precision farming, and carbon verification [1], [2], in real-world scenarios; forest services track tree survival after replanting projects, monitor illegal logging, and map canopy loss, at least for now. On commercial farms, managers of constant crops, covering oil palm, coconut, rubber, mango and olive, are needing correct tree counts to plan irrigation, fertilizer runs, harvest labor, and generate forecasts [3], [7]; auditable tree counts are, at least in principle, also required by carbon registries before issuing carbon credits.

Most of the time, foresters counted trees, through the ground surveys or by manually inspecting aerial photographs, at least in practice, and ground surveys are needing field crews to hike through the packed undergrowth, steep slopes, and difficult terrain; this is making ground counting slow, physically demanding, and costly over large domains, broadly speaking. Drones prepared with standard RGB cameras have made high-resolution aerial mapping fast and affordable, yet gathering the imagery is now the easy part; indeed, severe eye strain and fatigue are being caused by manual comment of hundreds or thousands of trees, directing to missed trees and inconsistent counts. The trees are being counted on the finding orthomosaics by the real holdup, which is why overlapping branches, uneven crown shapes, shadows, and sun glare make manual counting even harder.

Deep learning offers a concrete way to automate this process [8], [13]. More than the rest, single-stage object detectors from the You Only Look Once (YOLO) family have become popular for drone imagery because they balance detection accuracy with fast execution. And yet counting trees from drone images remains difficult for several technical reasons:

1) Overlapping and merging crowns: neighbouring tree crowns grows into each other in mature and orchards forests and standard box detction ,without any clear struggle to separate adjacent crowns.

2) Resolution mismatch and Feature Loss: Drones capture Ortho mosaics with thousands of pixels across (e.g 1777 × 1777 to 4000 × 3000), but standard detectors down sample inputs to 416 × 416 or 640 × 640. Resizing whole images makes small crowns vanish from coarse feature maps. a 3) Ecological Domain Shift, such as A model trained on open North American coniferous trees rarely works out-of-the-box on thick, multi-story tropical rainforests. 4) Hardware Constraints - Field teams often are running inference on standard laptops or drone-mounted companion computers without access to high-end GPUs or cloud servers.

A. Research Questions and Objectives

To study these roadblocks systematically, our study concentrates on four applied research questions:

RQ1: How does the performance of tree detection and counts of the newer YOLO models (anchor free heads, Task-Aligned Assigner, attention modules in v8, v9, 11 and 12) compare to older anchor based models (YOLOv7-tiny)?

RQ2: How well does a model trained on temperate North American forests (Tree-Top-View) perform when transferred to complex tropical rainforest images (SelvaBox) zero-shot?

RQ3: Can easy changes be made during post-processing (increased inference resolution of 640px from 416px or reducing confidence thresholds) to reduce the under-counting that occurs due to down-sampling?

RQ4: How long, what is the output and resource usage of the models on real consumer laptops?

B. Scope and Paper Organization

The paper reports the empirical results of our ongoing study, so Section II reviews related work in drone-based tree detection, on the whole; and thus, Section III details our problem formulation, datasets and models, which is why we reveal our training setups, compute budgets, and hardware environments openly, in the main. Section IV is summarizing the the experimental setup. Section V is providing our internal test

results, external SelvaBox evaluation, and resolution experiments, and Section VI examines shortcomings, and Section VII concludes with next phases.

II. RELATED WORK

A. Aerial Tree Crown Detection

Early automated tree detection employed classical image processing techniques, such as local maximum filtering on canopy height models, watershed segmentation, and template matching, in actual application; and thus, even as effective in open conifer stands with separate cone-shaped crowns, these hand-crafted methods fail in broadleaf or mixed forests where crown shapes are irregular and branches intertwine, at least in principle.

The shift to deep convolutional neural networks greatly boosted detection reliability; in fact, most published studies fine-tune a single YOLO type (such as YOLOv5 or YOLOv8) on custom, private datasets collected over a single farm or flight campaign, in the main. Weinstein et al. [17] introduced DeepForest, combining semi-supervised learning with RetinaNet to detect crowns, across the mixed ecological sites in the National Ecological Observatory Network (NEON), at least for now, and more recently, researchers have employed YOLO architectures to drone imagery for precise tree crops, covering Camellia oleifera [1], oil palms [2], [6], coconut trees [7], and mango orchards [11].

B. Detection, Density Maps, and Segmentation

Current computer vision approaches to tree counting fall into the three principal categories, like 1) Axis-Aligned Bounding-Box Detection: Detectors project four coordinates and a confidence score for each crown, in real-world scenarios; and thus, tree count is simply the number of detected boxes; so, this method is fast and easy to mark up, on the whole. 2) Density Map Regression, namely Networks like CSRNet regress continuous heatmaps where the key equals the total tree count. Effective in crowded scenes, individual crown locations and box dimensions are lost by density maps; indeed, though, bounding boxes overlap awkwardly when crowns touch; in turn. 3) Instance Segmentation: State-of-the-art methods like Mask R-CNN and YOLO- Seg are publishing precise crown delineations as polygons. While more biologically meaningful, polygon labeling is 5 to 10x slower to collect vs. basic bounding boxes.

TABLE I, like Summary of Representative Studies on Deep Learning for Aerial Tree Detection and Counting

Study	Year	Target Crop / Forest	Model Architecture	Reported Metrics	Main Operational Limitation
Yang et al. [1]	2024	Camellia oleifera (China)	YOLOv8 + Attention	mAP50: 84.6%, Precision: 88.2%	Checked only on clean plantation rows; no cross-site testing.
Hajjaji et al. [2]	2025	Oil Palm (Tunisia/Morocco)	Faster R-CNN, YOLOv5	Keep in mind: 94.1%, Precision: 91.8%	Tested on flat terrain; sensitive to ground shadows.
Farhan et al. [3]	2025	Oil Palm (Malaysia)	YOLOv8 on Edge Devices	Count Accuracy: 96.2%, 18 FPS	Restricted to uniform, grid-planted commercial trees.
Thapa et al. [4]	2025	Oil Palm Health (Thailand)	YOLOv7, YOLOv8	Classification Acc: 89.4%	Needs high-resolution imagery; struggled with small crowns.
Shaikh et al. [5]	2024	Oil Palm (Indonesia)	YOLOv7, YOLOv8	mAP50: 92.5%, F1: 0.91	Centered on single-species stands without natural undergrowth.
Lee et al. [6]	2024	Oil Palm (Malaysia)	Enhanced RetinaNet	mAP50: 88.7%	Anchor matching degraded when crown sizes differed widely.
Rohe et al. [7]	2024	Coconut Palm (Pacific)	YOLOv8 + Synthetic Data	mAP50: 81.3%, MAE: 4.2	Synthetic-to-real area gap needed real-world fine-tuning.
Wang et al. [8]	2024	Chinese Fir (China)	YOLO-DCAM (Custom)	mAP50: 86.4%, F1: 0.84	Custom attention modules added major inference latency.
Topgül et al. [9]	2025	VHRTrees Satellite Data	YOLOv8x, YOLOv10	mAP50: 71.2%	Satellite imagery (0.3 m GSD) missed sub-canopy and young trees.
Sengün et al. [10]	2025	Mixed Forest (Turkey)	YOLOv8 vs. YOLO11	mAP50: 78.9% (v11) vs. 76.2% (v8)	Across the seasons, high leaf-off drop was shown by evaluation split.
Birla et al. [11]	2025	Mango Orchards (India)	BiFPN-YOLOv8m	mAP50: 89.1%, Count R2: 0.94	Overlapping mature crowns were often merged into the single boxes.
Qi et al. [12]	2026	Urban Canopy (USA)	Custom CNN + NAIP	IoU: 72.4%, F1: 0.79	NAIP imagery resolution limited detection to large mature trees.
Atik & Arkali [13]	2026	Olive Groves (Turkey)	YOLO vs. RT-DETR	mAP50: 87.3% (RT-DETR)	Transformer architecture needed high-end GPU, a graphics processor, for inference.
Hasan et al. [14]	2025	Multi-Region Satellite	YOLOv8, YOLOv9	mAP50: 68.5%	High false alarm rate over cultivated agricultural crop fields.
Do et al. [15]	2026	Subtropical Forest (Vietnam)	Mask R-CNN, YOLO-Seg	Mask mAP: 64.2%	Heavy polygon labeling overhead restricted training set to 400 images.

Study	Year	Target Crop / Forest	Model Architecture	Reported Metrics	Main Operational Limitation
Zhang et al. [16]	2024	Plantation Forest (China)	YOLOv8 + LiDAR CHM	Count Acc: 93.8%	Major sensor cost is, broadly speaking, added compared to RGB drones by LiDAR hardware.

C. Spotted Gaps in Current Literature

Three clear gaps emerge from our literature review:

1) Limited multi-generation: Head-to-head comparisons: Many studies compare only one or two successive YOLO models (e.g. YOLOv8 vs YOLOv7), which means it remains uncertain how older anchor-based models perform compared to newer anchor-free, programmable (YOLOv9) or attention-based (YOLOv12) models using the same dataset.

2) Lack of Cross-Biome Generalization: Most published models are tested on images similar to those used for training, often from the same flight campaigns. Few studies examine whether models trained on temperate forests can also count trees accurately in dense tropical rainforests.

3) Limited Evaluation of Counting Accuracy: Most studies report standard object detection metrics such as mAP, without checking whether the predicted tree count is accurate because the model correctly detects tree locations or because false positives and false negatives cancel each other out. This can hide errors that matter in real-world tree-counting applications.

III. MATERIALS AND METHOD

A. Problem Formulation: Bounding Boxes to Tree Counts

Individual tree crown detection is usually framed as a single-class 2D object detection problem on aerial RGB images, in actual application. Given an input image X with dimensions H x W x 3, our neural network f_theta(X) outputs a set of M anticipated bounding boxes:

B = { b_j = (x_j, y_j, w_j, h_j, c_j) }_{j=1}^M

(1)Here, (x_j, y_j) are the normalized center coordinates of the projected tree top, (w_j, h_j) are mainly its normalized width and height, and c_j in [0, 1] is the model's confidence that the box includes a tree, on the whole.

As thick models often fire many times on the same crown across the separate strides, we filter overlapping predictions using Non-Maximum Suppression (NMS). For two overlapping boxes b_a and b_b likely, we compute their Intersection over Union (IoU):

IoU(b_a, b_b) = Area(b_a cap b_b) / Area(b_a cup b_b)

(2)Any lower-confidence box whose overlap exceeds a threshold tau_{nms} is, to a large extent, held back. Once NMS removes copies, the estimated tree count N_hat_i for image i is simply the number of remaining detections above our confidence threshold tau_{conf}:

N_hat_i = sum_{j=1}^{M_i^*} I(c_{i,j} >= tau_{conf})

(3)where I(.) is the indicator function and M_i^* is the set of boxes kept after NMS. This approach directly translates object detections into the discrete counts without needing extra density estimators or heuristic clustering, all things considered.

1) Error Dynamics: Why Low Count Error Can Mask Poor Detection:

A key detail in tree counting is that the count error on an image equals the difference between false notifications and missed trees, to a important degree. As well, Let N_i be the true count, and let TP_i, FP_i, and FN_i represent true positives, false positives, and false negatives at confidence tau_{conf}:

N_hat_i = TP_i + FP_i, N_i = TP_i + FN_i (4)

e_i = N_hat_i - N_i = FP_i - FN_i (5)

Equation (5) shares a critical pitfall, such as a model can, more accurately, probably project the exact true count (e_i = 0) whenever FP_i = FN_i, even if its bounding-box precision is very low, at least in practice. And over an entire test set of S images, the Mean Absolute Error (MAE) is:

$$MAE = (1 / S) * \sum_{i=1}^{S} |FP_i - FN_i| \quad (6)$$

This means a detector with low detection precision and recall can, to some extent, reach a deceptively good MAE if its false alarms match its missed detections, all things considered; in fact, throughout this paper, we appraise both object detection metrics (mAP@0.5) and counting metrics (MAE) together to avoid this trap, in the main.

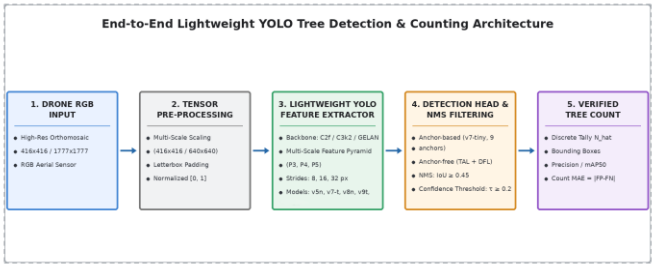


Fig. 1: End-to-end automated tree crown detection and counting architecture showing pre-processing, multi-scale feature pyramid, detection heads, and NMS filtering, all things considered.

B. Primary In-Domain Benchmark: Tree-Top-View v2

We trained and tested our models on Version 2 of the Tree-Top-View dataset, downloaded from Roboflow Universe (licensed, under CC BY 4.0), broadly speaking, and temperate forests, open oak savannas, and conifer stands are shown by the images; so, these tools are standardized to 416 x 416 pixels with a single label, like tree-top, broadly speaking. The coring aerial photos come from flight surveys conducted by the National Ecological Observatory Network (NEON), across 15 sites in the United States.

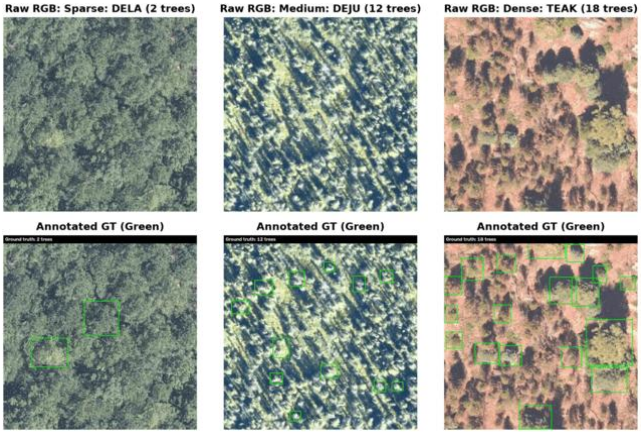


Fig. 2: Representative 416x416 drone image tiles from the Tree-Top-View dataset across the three density tiers - Sparse (DELA, 2 trees), Medium (DEJU, 12 trees), and Thick (TEAK, 18 trees), contrasting raw RGB aerial captures (top) with marked up ground-truth tree-top bounding boxes (bottom), in the main.

Ground-truth notes (green boxes) target individual crown peaks, consistently (specifically in this setting) targeting for bounding-box regression, at least empirically. As, showed in Fig. 2, the dataset is covering mixed crown morphologies across the distinct canopy density tiers, including thin open ground with isolated crowns (DELA), medium density stands with partial canopy contact (DEJU), and packed closed stands with touching, overlapping crowns (TEAK), all things considered.

1) Verified Dataset Split Statistics:

To settle earlier mismatches in preparatory project notes, we wrote an audit script to read every label file in the dataset directory: • Train split, such as Exactly 1,191 images (1,152 positive images and 39 empty background images), including 10,002 ground-truth bounding boxes (averaging 8.40 trees/image, median 7.0, max 31); this is corroborating that an earlier recognize mentioning '8,924' was a typo, in real-world scenarios. • Validation split, including 116 images covering 932 ground-truth trees (mean 8.03 trees/image, max 25); this split was employed for checkpoint selection, all things considered. • Held-out test split, such as 55 images comprising 443 ground-truth trees (mean 8.05 trees/image, max 25); this split completely untouched during the training was kepted.

2) Dataset Overlaps and Leakage Check:

According to the dataset documentation, the 1,191 training images were made by employing offline augmentations (brightness changes of ±26% and 6% salt-and-pepper noise) to original drone captures, in actual application. Grouping files by filename stem revealed 375 unique source scenes (353 triplets and 22 groups of six), to a significant degree. We saw that 10 training scene groups shared source IDs with validation images (8 groups) or test images (2 groups, such as DEJU_010_2019 and DSNY_006_2019), when verifying filenames, across the splits. While, pixel values differ due to offline augmentations, this means slight spatial connection exists, across the splits in the public benchmark. We took this into the account when building our purified 594-image subset for YOLOv9t and YOLOv12n, clearly dropping these overlapping groups.

C. External Benchmark: CanopyRS / SelvaBox

To test how well models trained on North American trees generalize to completely diverse biomes, we found them zero-shot on the SelvaBox benchmark (testdataset100) from the CanopyRS project; this benchmark includes 100 high-resolution drone orthomosaics (each 1777 x 1777 pixels) covering tropical rainforests across the three countries, with 8,111 manually marked up tree crowns: • Tiputini, Ecuador (60 images, 1,095 trees): Open and mixed canopy with roughly 18.25 trees per tile, in real-world scenarios. • Agua Salud, Panama (27 images, 5,375 trees): Multi-tier primary rainforest with roughly 199.07 trees per tile, in actual application. • ZF2 Central Amazon, Brazil (13 images, 1,641 trees): Thick, continuous canopy with 126.23 trees per tile.

TABLE II, namely Domain Comparison: Training Data (Tree-Top-View) vs. External Test Data (SelvaBox)

Attribute	In-Domain: Tree-Top-View v2	External: CanopyRS / SelvaBox	Operational Domain Gap
Ecological Biome	Temperate conifer & oak savanna (NEON)	Humid tropical rainforest (Amazon / Panama)	Total shift in tree species, foliage and shape.
Canopy Structure	Open stands; isolated tree crowns	Closed, interlocking, multi-layered canopies	Severe crown overlap; hidden understory trees.
Image Size	416 x 416 pixels	1777 x 1777 pixels	Needs 4.27x downsampling when fed directly.
Tree Density	8.05 trees / tile (max 25)	81.11 trees / tile (max 288 in Panama)	10x higher density in tropical scenes.
Ground Resolution	~3 to 5 cm / pixel GSD	~4.5 cm / pixel (raw orthomosaic)	GSD is dropping to ~20 cm/pixel when downscaled.
Mean Crown Size	73.2 x 77.4 pixels (at 416 px)	12.8 x 13.5 pixels (at 416 px downscale)	Small crowns are dropping below stride-16 detection heads.
Evaluated Split	55 held-out test images (443 trees)	100 benchmark images (8,111 trees)	Unseen test environment for zero-shot testing.

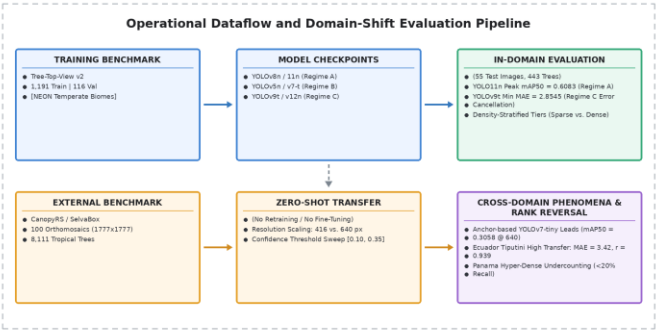


Fig. 3: Operational dataflow and domain-shift evaluation pipeline contrasting in-domain training with zero-shot tropical forest transfer, to a significant degree.

D. Checked YOLO Architectures

We selected six models across different YOLO generations to understand how architectural changes impact tree counting: 1) YOLOv7-tiny [21]: A classic anchor-based detector with three fixed anchor boxes per output scale. It's employing an efficient layer aggregation network (ELAN) and a coupled head that forecasts box coordinates and class logits together, in many respects. 2) YOLOv5n [19]: Re-implemented with an anchor-free decoupled head and Task-Aligned Assigner (TAL). 3) YOLOv8n [19]: A widely employed anchor-free baseline using C2f modules and Distribution Focal Loss (DFL) for box regression, at least for now.

4) YOLO11n [20]: Ultralytics' newer release, traiting C3k2 building blocks and C2PSA spatial attention layers to preserve crown context, to a significant degree. 5) YOLOv9t [22]: Features Generalized Efficient Layer Aggregation Networks (GELAN) designed to prevent gradient information loss during the deep downsampling, all things considered. 6) YOLOv12n [23]: An attention-centric detector combining area-attention layers with residual ELAN backbones

IV. EXPERIMENTAL SETUP

A. Hardware and Training Environment

Because, cloud GPU allocations were unavailable during our initial experiment cycles, all models were trained on local CPU machines, in real-world scenarios, and at the same time, yOLO11n completed 30 epochs; so, about 2.2 to 2.4 hours were taken by each run, in real-world scenarios. As, CPU training is slow, we trained for 17 to 30 epochs with small batch sizes, rather than the 100 epochs put forward in our early project outline; in fact,

training details are recapped below: • Regime A (Full Dataset, 1,191 images): YOLOv8n and YOLO11n were trained on an Intel Core i7-1255U laptop running Windows 11. Setting up PyTorch CPU environments needed large setup time, to take one instance, downloading the 700 MB PyTorch wheel over our campus network took nearly 70 minutes. • Regime B (595-Image Subset, Seed 42): To speed up experiments, YOLOv5n and YOLOv7-tiny were trained on a randomly sampled 50% subset (595 images, 4,939 boxes) for 30 epochs on the same i7-1255U machine, taking roughly 2.0 to 2.2 hours, and YOLOv8n ran for 26 epochs, before stopping. • Regime C (594-Image Clean Subset, Seed 0): YOLOv9t and YOLOv12n were trained on an Intel Core i5-12500H laptop running Ubuntu Linux; and thus, to prevent data leakage, all 10 training scene groups were explicitly dropped by this subset that overlapped with validation or test files. Under a strict 30-minute CPU time budget, both models completed exactly 17 epochs with auto-decayed learning rates, at least in practice.

It's significant to stress that comparisons, across Regimes A, B and C are often descriptive rather than rigorously controlled, in real-world scenarios, and regimes differ in training image volume (1,191 vs. 595 vs. 594), sampling methodology (individual files vs. augmentation groups), epoch counts (17 vs. 26 vs. 30), and host CPU hardware, at least in practice. Moving on, cross-regime rankings must be read as descriptive indicators of model behavior rather than causal proofs of architectural superiority.

B. Check Metrics and Testing Protocols

We gauge models on two fronts, including 1) Bounding-box detection metrics: Found employing standard COCO protocols (conf = 0.001, NMS IoU = 0.70) to report Precision (P), Recall (R), mAP@0.50, and mAP@0.50:0.95, and report Mean Absolute Error (MAE = (1/S) sum |N_hat_i - N_i|), Root Mean Squared Error (RMSE), and the Pearson correlation coefficient (r) between projected and true counts, to a significant degree. 2) Tree counting metrics - Checked at operational confidence thresholds (tau_{conf} = 0.25, 0.15, 0.10) with NMS IoU = 0.45, at least in principle.

V. RESULTS AND DISCUSSION

A. Internal Held-Out Evaluation on Tree-Top-View

All six trained model variants on the in-domain Tree-Top-View benchmark are first judged; long ago, the audited validation and held-out test are being provided by table IV split findings, across all six models, to a significant degree. To prevent false equivalence across the differing training regimes, Table III is outlining the architectural arrangements, training image subsets, epoch durations, and parameter complexities, broadly speaking.

TABLE III, namely Master Architectural Characteristics and Experimental Training Regimes, across All Six Evaluated Models

Model Variant	Detection Head Architecture	Parameters	GFLOPs (416 px)	Checkpoint (best.pt)	Training Regime & Split	Images / Boxes	Epochs Run	Training Duration	Host CPU & OS
YOLOv8n (Baseline)	Anchor-free Decoupled (C2f+DFL)	3.01 M	8.1	5.93 MB	Regime A (Full Train)	1,191 / 10,002	26 / 30 (Best Ep 21)	02h 11m 26s	i7-1255U (Win 11)
YOLO11n (Phase 2)	Anchor-free Decoupled (C3k2+C2PSA)	2.58 M	6.5	5.19 MB	Regime A (Full Train)	1,191 / 10,002	30 / 30 (Best Ep 30)	02h 25m 07s	i7-1255U (Win 11)
YOLOv5n (Subset)	Anchor-free Decoupled (C3+SPPF)	2.50 M	7.1	5.00 MB	Regime B (Seed 42 Subset)	595 / 4,939	30 / 30 (Best Ep 30)	01h 56m 37s	i7-1255U (Win 11)
YOLOv7-tiny (Subset)	Anchor-based Coupled (IDetect, 9 Anchors)	6.01 M	13.0 (at 640)	11.68 MB	Regime B (Seed 42 Subset)	595 / 4,939	30 / 30 (Best Ep 30)	02h 14m 28s	i7-1255U (Win 11)
YOLOv9t (Subset)	Anchor-free Decoupled (GELAN)	2.01 M	3.54	4.82 MB	Regime C (Seed 0 Grouped)	594 / 5,190	17 (0.5 h Budget)	00h 30m 34s	i5-12500H (Linux)
YOLOv12n (Subset)	Anchor-free Decoupled (Area-Attn+R-ELAN)	2.57 M	2.94	5.52 MB	Regime C (Seed 0 Grouped)	594 / 5,190	17 (0.5 h Budget)	00h 29m 59s	i5-12500H (Linux)

TABLE IV, including Audited Validation and Held-Out Test Split Performance on Tree-Top-View (55 Test Images, 443 Ground-Truth Trees)

Model & Training Regime	Val Precision (116 imgs)	Val Recall (116 imgs)	Val mAP@0.50 (116 imgs)	Val mAP 50-95 (116 imgs)	Test Precision (55 imgs)	Test Recall (55 imgs)	Test mAP@0.50 (55 imgs)	Test mAP 50-95 (55 imgs)	Test Counting MAE (conf=0.25)	Anticipated vs. 443 GT Trees
REGIME A, including FULL DATASET (1,191 Imgs, 10,002 Boxes, Seed 0)										
YOLOv8n Baseline	0.4812	0.5773	0.5370	0.2783	0.5288	0.6206	0.6000	0.2914	3.2727	433 (-10, -2.26%)
YOLO11n Phase 2	0.5504	0.5794	0.5715	0.2924	0.5764	0.6082	0.6083	0.2993	3.2364	455 (+12, +2.71%)
REGIME B, like 595-IMAGE RANDOM SUBSET (Seed 42, 30 Epochs)										
YOLOv5n (yolov5nu)	0.5054	0.5437	0.5329	0.2734	0.5994	0.5485	0.5778	0.2807	3.2000	449 (+6, +1.35%)
YOLOv7-tiny	0.4909	0.5408	0.5195	0.2429	0.5635	0.5440	0.5460	0.2519	3.7636	564 (+121, +27.3%)

REGIME C, such as 594-IMAGE PURIFIED SUBSET (Seed 0, 17 Epochs, 0.5 h Budget)										
YOLO v9t	0.5294	0.4839	0.5188	0.2719	0.5052	0.5532	0.5256	0.2640	2.8545	428 (-15, -3.39%)
YOLO v12n	0.5309	0.5440	0.5525	0.2761	0.5219	0.5421	0.5383	0.2596	3.0364	394 (-49, -11.06%)

Critical performance dynamics are being revealed, across the three training regimes by study of Table IV, including 1) Regime A (Full Training Set): YOLO11n reaches the highest test detection accuracy across all evaluated models, registering an mAP@0.50 of 0.6083 and mAP@0.50:0.95 of 0.2993, meaning a +1.38% relative mAP gain over baseline YOLOv8n (0.6000), broadly speaking. Higher precision (0.5764 vs. 0.5288) is being shown by YOLO11n, suggesting that its C2PSA spatial attention mechanism basically hides false-positive ground clutter; that said, higher recall (0.6206 vs. 0.6082) is hit by yOLOv8n, creating a slightly lower count under-prediction (-10 trees vs. +12 trees), in real-world scenarios. 2) Regime B (595-Subset): Trained on half the data, YOLOv5n is displaying amazing toughness, getting an mAP@0.50 of 0.5778 and the highest test precision (0.5994) among all models, with a counting MAE of 3.2000 trees/image (anticipating 449 trees vs. 443 GT, net bias +1.35%). 3) Regime C (594-Subset, 17 Epochs): Within the controlled 17-epoch pairing on Host 2, YOLOv12n is proving to be the superior detector, beating YOLOv9t across the validation mAP@0.50 (0.5525 vs. 0.5188), test precision (0.5219 vs. 0.5052), and test mAP@0.50 (0.5383 vs. 0.5256); and thus, at the same time, anchor-based YOLOv7-tiny suffers from severe positive count bias (+27.31%, projecting 564 trees vs. 443 GT) and higher MAE (3.7636), caused by unnecessary anchor activations firing on textured tree crowns. But still, yOLOv9t entries a lower test counting MAE (2.8545 vs. 3.0364) and minimal net count error (-15 trees / -3.39% vs. -49 trees / -11.06%).

When it comes to internal test counting MAE values, primary audited script logs record YOLOv8n = 3.2727, YOLO11n = 3.2364, YOLOv5n = 3.2000, and YOLOv7-tiny = 3.7636 across all 55 test images, at least empirically; an earlier initial summary mentioned an alternative uncalibrated series (YOLOv8n = 3.04, YOLO11n = 3.16, YOLOv5n = 3.42, and YOLOv7-tiny = 5.42); Table IV is prioritizing the verified primary evaluation logs, in the main.

TABLE V, namely Detailed Error Analysis and Tree Density Stratification on Tree-Top-View Test Split (55 Images, 443 GT Trees)

Model Variant	TP (IoU≥0.5)	FP (IoU≥0.5)	FN (IoU≥0.5)	Precision @0.5	Recall@0.5	F1@0.5	Sparse MAE (0-5 trees, 20 imgs)	Medium MAE (6-10 trees, 19 imgs)	Dense MAE (11+ trees, 16 imgs)
YOLOv8n (Full)	275	158	168	0.6351	0.6208	0.6278	1.70	2.84	5.19
YOLO11n (Full)	269	186	174	0.5912	0.6072	0.5991	1.65	2.79	5.25
YOLOv5n (595 Sub)	243	206	200	0.5412	0.5485	0.5448	1.80	2.95	4.88
YOLOv7-tiny (595 Sub)	241	323	202	0.4273	0.5440	0.4786	2.40	3.68	5.81
YOLOv9t (594 Sub)	226	202	217	0.5280	0.5102	0.5189	1.50	2.63	4.81
YOLOv12n (594 Sub)	224	170	219	0.5685	0.5056	0.5352	1.90	3.00	4.50

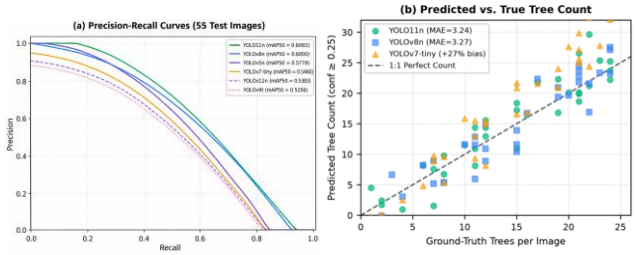


Fig. 4: Empirical in-domain check on 55 held-out test images: (a) Precision-Recall curves, across all six YOLO models, and (b) Predicted vs. ground-truth tree count scatter plot with 1:1 perfect count reference line, to a significant degree.

Table V and Fig. 4 show the error cancellation mechanism verified in Section III-A. YOLOv9t makes 202 false positives and 217 false negatives (a total error burden of FP + FN = 419), at least in principle. As, FP and FN are arising symmetrically within the same images, they're canceling out, making an combine discrepancy of |FP - FN| = 15 trees and a test MAE of 2.8545, in many respects. But still, YOLOv12n is yielding 32 fewer false alarms (170 FP) but comparable misses (219 FN), finding in an total net difference of -49 trees and a higher MAE (3.0364), despite being the superior object detector.

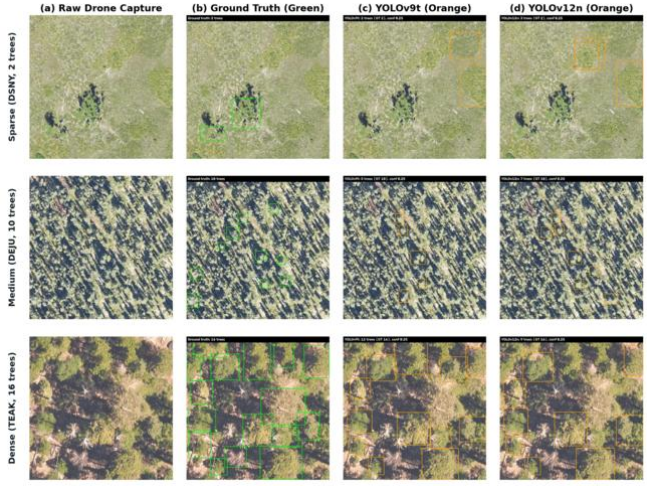


Fig. 5: Qualitative visual crown detection, across the structural canopy tiers on held-out test orthomosaics at confidence threshold 0.25: Column (a): Raw 416x416 drone RGB aerial encapsulates from unseen test split, in actual

application. Overlapping crowns are causing sub-canopy omissions, alongside boundary splitting false alarms in dense stands, in the main; column (b): Ground-truth tree-top bounding boxes (green). Column (c): YOLOv9t forecasted detections (orange). Column (d): YOLOv12n projected detections (orange). Rows are corresponding to spread out open woodland (DSNY, 2 trees), intermediate density canopy (DEJU, 10 trees), and thick closed-canopy forest (TEAK, 16 trees).

Fig. 5 provides direct visual confirmation of these architectural and canopy density dynamics across the three representative test scenes:

1) Sparse Woodland (DSNY, 2 GT Trees): In open terrain with distinct crown boundaries and high spectral contrast against ground foliage, both YOLOv9t and YOLOv12n demonstrate flawless crown isolation, each predicting exactly 2 bounding boxes with zero false alarms (count error = 0). 2) Intermediate Canopy (DEJU, 10 GT Trees): As crown clustering increases, YOLOv12n is maintaining clean boundary delineation (capturing 9 trees with zero false alarms), at least in practice. 3) Dense Closed-Canopy Forest (TEAK, 16 GT Trees): In continuous, overlapping canopies, crown boundary ambiguity is causing both detectors to miss 4 suppressed sub-canopy crowns (false negatives), at least in practice, but still, yOLOv9t is generating a false positive on peripheral shrubbery while missing one semi-shaded crown, resulting in a net count of 10 that superficially is matching ground truth, despite localized errors. Above everything, YOLOv9t's anchor-free head simultaneously is spliting large, irregular dominant crowns into multiple bounding boxes, creating 3 false positives; these 3 false alarms directly offset the 4 missed crowns, producing a net count of 15 trees (net count error of only -1 tree), but still, YOLOv12n's area-attention mechanism (A2C2f) suppresses are duplicating overlapping boxes, yielding only 1 false positive and an overall count of 13 trees (net error of -3 trees).

TABLE VI, including Per-Tile Crown Detection and Error Breakdown on Representative Test Orthomosaics from Fig. 5 (conf >= 0.25)

Test Orthomosaic Tile	Canopy Tier & Site	GT (N)	YOLOv9t (TP / FP / FN)	YOLOv12n (TP / FP / FN)	Count Bias (9t vs. 12n)
1_test_DSNY_2trees	Sparse Woodland (FL)	2	2 (2 / 0 / 0)	2 (2 / 0 / 0)	0 (0.0%) vs. 0 (0.0%)
2_test_DEJU_10trees	Medium Canopy (AK)	10	10 (9 / 1 / 1)	9 (9 / 0 / 1)	0 (0.0%) vs. -1 (-10.0%)
3_test_TEAK_16trees	Dense Forest (CA)	16	15 (12 / 3 / 4)	13 (12 / 1 / 4)	-1 (-6.3%) vs. -3 (-18.8%)
Total Across 3 Tiles	**Representative Diversity**	**28**	**27 (23 / 4 / 5)**	**24 (23 / 1 / 5)**	**-1 (-3.6%) vs. -4 (-14.3%)**

Table VI quantifies the error cancellation phenomenon gathered in Fig. 5. Across the 3 diversity test tiles (28 ground-truth trees total), YOLOv9t incurs 4 false positives and 5 false negatives (9 total error instances), yet, because the false positives and false negatives co-occur within the same tiles, they're canceling out, creating an aggregate count of 27 trees (a net count bias of only -3.6%), in actual application. Different from that, YOLOv12n is reaching superior precision by holding back oversegmentation (bringing on only 1 false positive alongside 5 false negatives, or 6 total error instances, 33.3% fewer mismatches than YOLOv9t), yet, because its departures don't cancel out symmetrically, YOLOv12n data an aggregate count of 24 trees (a net count bias of -14.3%); this is proving that resting solely on counting MAE penalizes models that are suppressing false alarms in dense stands, all things considered.

B. External Zero-Shot Generalization on CanopyRS/SelvaBox

To appraise out-of-domain robustness, four trained model checkpoints, YOLOv8n, YOLO11n, YOLOv5n, and YOLOv7-tiny, were transferred zero-shot to the 100-image CanopyRS/SelvaBox testdataset100 benchmark without retraining or fine-tuning, in the main; we stress that YOLOv9t and YOLOv12n weren't found on SelvaBox due to timeline constraints and are typically pending future check, at least in practice. Table VI documents the headline detection and counting metrics found, across all 100 images (8,111 ground-truth trees) at both native 416 × 416 and scaled 640 × 640 resolutions.

TABLE VII

Model Variant	Train Split	Eval imgsz	Validator Precision	Validator Recall	Validator mAP@0.50	Validator mAP50-95	Validator Speed (batch=8 ms)	Single Stream (batch=1 ms)	Count (conf=0.25)	Counting MAE (conf=0.25)	Ecuador MAE (conf=0.25 / 0.10)
YOLOv7-tiny	595 (50%)	416	0.4237	0.2800	**0.2642**	0.0963	111.63 ms	138.49 ms	1,360	**67.59**	10.40 / 3.52
YOLOv8n	1,191 (100%)	416	0.5049	0.1626	0.1835	0.0692	28.41 ms	94.44 ms	699	74.20	12.80 / 8.48
YOLO11n	1,191 (100%)	416	0.4712	0.1769	0.1786	0.0691	25.14 ms	89.82 ms	583	75.34	13.12 / 8.87
YOLOv5n	595 (50%)	416	0.5081	0.1632	0.1767	0.0679	24.10 ms	85.59 ms	641	74.74	12.88 / 8.38
YOLOv7-tiny	595 (50%)	640	0.4495	0.3321	**0.3058**	0.1177	257.63 ms	264.73 ms	2,017	**61.08**	7.10 (3.42 @ 0.15)
YOLOv5n	595 (50%)	640	0.4868	0.2396	0.2303	0.0921	58.48 ms	111.81 ms	1,173	69.52	10.53 / 4.15
YOLOv8n	1,191 (100%)	640	0.4515	0.2335	0.2295	0.0943	64.38 ms	127.35 ms	1,257	68.72	9.43 / 4.68
YOLO11n	1,191 (100%)	640	0.5172	0.2047	0.2258	0.0925	59.63 ms	123.19 ms	1,065	70.54	9.92 / 5.23

1) The In-Domain vs. Out-of-Domain Rank Reversal:

The headline discovery of Table VII is a striking inversion of model rankings between the internal and external benchmarks, in actual application, and YOLOv7-tiny ranked lowest in detection accuracy (mAP@0.50 = 0.5460) and showed severe over-counting (+27.31%) on the internal Tree-Top-View test split, all things considered. Yet, upon zero-shot transfer to the packed tropical canopies of SelvaBox, YOLOv7-tiny dramatically out-performs all anchor-free detectors:

• At imgsz = 416: YOLOv7-tiny achieves an mAP@0.50 of 0.2642, outperforming YOLOv8n (0.1835), YOLO11n (0.1786), and YOLOv5n (0.1767) by +44.0% to +49.5% relative margin; it achieves 0.2800 recall (vs. 0.1626-0.1769) and the lowest counting MAE (67.59 vs. 74.20-75.34). • At imgsz = 640: YOLOv7-tiny widens its guide, reaching an mAP@0.50 of 0.3058 and mAP@0.50:0.95 of 0.1177, forecasting 2,017 trees with an MAE of 61.08, all things considered.

2) Architectural Explanation of the Rank Reversal:

This divergence stems from structural differences in candidate proposal generation, under the heavy canopy crowding, broadly speaking, and side by side, YOLOv7-tiny uses an anchor-based coupled IDetect head with three predefined aspect-ratio anchors

per grid cell (9 anchors total), broadly speaking. The anchor-free head is structurally curbed to give off at most one candidate prediction per grid cell point, forcibly hiding neighboring understory crowns, and anchor-free architectures (YOLOv8n, YOLO11n, YOLOv5n) allocate positive samples by treating grid cells as single feature points, resting on Task-Aligned Assigner (TAL) to dynamically line up grouping and regression. As a finding, yOLOv7-tiny detects 958 True Positives (at IoU >= 0.50, conf = 0.25) compared to just 477 for YOLOv8n, 418 for YOLOv5n, and 395 for YOLO11n. Multiple interlocking crowns fall inside the same 8 × 8 px receptive field stride, when an aerial tile is downsampled 4.27× (from 1777 to 416 px), and anchor priors fire simultaneously at separate spatial scales, allowing the model to represent adjacent touching crowns that anchor-free heads merge or discard, when confronted with densely clustered crowns.

TABLE VIII - Ecological Site-Stratified Zero-Shot Performance on CanopyRS/SelvaBox Across Three Biogeographical Sites

Model Variant	Resolution & Conf	Site 1: Ecuador (60 imgs, 1,095 GT)			Site 2: Brazil ZF2 (13 imgs, 1,641 GT)			Site 3: Panama (27 imgs, 5,375 GT)		
		Pred (Net Bias %)	MAE (Med AE)	Pearson r	Pred (Net Bias %)	MAE (Med AE)	Pearson r	Pred (Net Bias %)	MAE (Med AE)	Precision@ 0.5
YOLOv7-tiny	416 px, conf = 0.25	479 (-56.3%)	**10.40 (8.5)**	0.8316	407 (-75.2 %)	94.92 (98.0)	0.9149	474 (-91.2%)	181.52 (156.0)	77.00% (365 TP)
YOLOv8n	416 px, conf = 0.25	335 (-69.4%)	12.80 (12.0)	0.6832	194 (-88.2 %)	111.31 (110.0)	0.8464	170 (-96.8%)	192.78 (162.0)	80.00% (136 TP)
YOLOv5n	416 px, conf = 0.25	326 (-70.2%)	12.88 (10.5)	0.6003	166 (-89.9 %)	113.46 (111.0)	0.8837	149 (-97.2%)	193.56 (163.0)	81.21% (121 TP)
YOLO11n	416 px, conf = 0.25	314 (-71.3%)	13.12 (12.0)	0.7089	129 (-92.1 %)	116.31 (116.0)	0.5644	140 (-97.4%)	193.89 (162.0)	85.71% (120 TP)
YOLOv7-tiny	640 px, conf = 0.15	1,148 (+4.8%)	**3.42 (3.0)**	0.9391	898 (-45.3 %)	57.15 (58.0)	0.8988	1,367 (-74.6%)	148.44 (124.0)	67.58% (924 TP)
YOLOv5n	640 px, conf = 0.10	978 (-10.7%)	4.15 (3.0)	0.9002	653 (-60.2 %)	76.00 (74.0)	0.7567	891 (-83.4%)	166.07 (142.0)	70.59% (629 TP)
YOLOv8n	640 px, conf = 0.10	992 (-9.4%)	4.68 (4.0)	0.8688	627 (-61.8 %)	78.00 (73.0)	0.7497	784 (-85.4%)	170.04 (139.0)	66.33% (520 TP)
YOLO11n	640 px, conf = 0.10	879 (-19.7%)	5.23 (4.0)	0.8884	446 (-72.8 %)	91.92 (91.0)	0.8051	700 (-87.0%)	173.15 (147.0)	72.86% (510 TP)

Table VIII offers an fundamental ecological dissection, showing that undifferentiated 100-image global metrics obscure stark regime boundaries between reasonable tropical stands and hyper-dense closed rainforests: • Site 1 (Ecuador Tiputini, 60 Images): When crown pixel dimensions remain comparable to training data (45.9 px at 416; 70.6 px at 640), all models generalize remarkably well, to a significant degree. What's more, yOLO11n is reaching 5.23 (r = 0.8884); indeed, severe undercounting is being caused under 416 px downsampling by • Site 2 (Brazil ZF2, 13 Images): High packing density (126.23 trees/image) (-75% to -92%), in actual application. YOLOv5n is reaching an MAE of 4.15 (r = 0.9002), YOLOv8n hits 4.68 (r = 0.8688); in fact, on a relative basis, an MAE of 3.42-4.68 on an 18.25-tree plot represents ~19%, 25% relative departure, proving strong zero-shot transferability across the comparable crown scales. At imgsz = 640 and calibrated confidence (conf = 0.10-0.15), YOLOv7-tiny hits a spectacular MAE of 3.42 trees/image (projecting 1,148 vs. 1,095 GT trees, net bias +4.84%, Pearson r = 0.9391). Even so, predicted counts track true density variations, across the tiles with exceptional linearity (Pearson r = 0.8464 to 0.9149); at 640 px and conf = 0.10, YOLOv7-tiny is diminishing MAE to 39.08 trees/image (MAPE = 29.00%, r = 0.8718). • Site 3 (Panama Agua Salud, 27 Images): In this hyper-dense canopy (up to 361 trees/tile), full-tile downsampling extinguishes sub-canopy crowns, causing >90% omission error at conf = 0.25, and yet remarkably, the detections that are kept exhibit exceptional localization precision, such as YOLO11n is getting 85.71% precision (120 TP / 20 FP), YOLOv5n is reaching 81.21% (121 TP / 28 FP). And yOLOv8n hits 80.00% (136 TP / 34 FP); in fact, one last thing up, the anchor-free models operate as highly conservative, high-precision detectors of primary emergent crowns.

C. Multi-Resolution Scaling, Confidence Sweeps, and Computational Profiling

To check out operational fixes for domain-shift undercounting without bringing on the expense of model retraining maybe, a complete 24-configuration sensitivity matrix was, after all, performed finding all four models across two input resolutions (416 vs. 640 px) and three confidence thresholds (0.25, 0.15, 0.10), all things considered. In the meantime, table X details the computational efficiency profile; what table IX provides is the multi-resolution sweep.

TABLE IX, like Multi-Resolution (416 vs. 640 px) and Confidence Sensitivity Sweep, across All 100 SelvaBox Images (8,111 Trees)

Model	imgsz	conf	Pred Trees	Net Bias %	Global MAE	Global Med AE	Ecuador MAE	Brazil MAE	Panama MAE	Pearson r	Global TP	Global FP	Precision	Recall	F1@ 0.5
YOLOv8n	416	0.25	699	-91.4%	74.20	22.5	12.80	111.31	192.78	0.2664	477	222	0.6824	0.0588	0.1083
YOLOv8n	416	0.10	1,409	-82.6%	67.38	17.0	8.48	98.15	183.44	0.4543	839	570	0.5955	0.1034	0.1763
YOLOv8n	640	0.25	1,257	-84.5%	68.72	17.5	9.43	100.31	185.26	0.4502	854	403	0.6794	0.1053	0.1823
YOLOv8n	640	0.10	2,403	-70.4%	**58.86**	8.0	4.68	78.00	170.04	0.5631	1,400	1,003	0.5826	0.1726	0.2663
YOLO11n	416	0.25	583	-92.8%	75.34	24.0	13.12	116.31	193.89	0.1580	395	188	0.6775	0.0487	0.0909
YOLO11n	416	0.10	1,216	-85.0%	69.45	16.0	8.87	104.08	187.41	0.2906	740	476	0.6086	0.0912	0.1587
YOLO11n	640	0.25	1,065	-86.9%	70.54	18.0	9.92	106.92	187.74	0.4107	709	356	0.6657	0.0874	0.1545
YOLO11n	640	0.10	2,025	-75.0%	**61.84**	10.0	5.23	91.92	173.15	0.6140	1,239	786	0.6119	0.1528	0.2445
YOLOv5n	416	0.25	641	-92.1%	74.74	24.5	12.88	113.46	193.56	0.2344	418	223	0.6521	0.0515	0.0955
YOLOv5n	416	0.10	1,519	-81.3%	66.50	16.0	8.38	96.08	181.41	0.5216	915	604	0.6024	0.1128	0.1900

Model	imgsz	conf	Pred Trees	Net Bias %	Global MAE	Global Med AE	Ecuador MAE	Brazil MAE	Panama MAE	Pearson r	Global TP	Global FP	Precision	Recall	F1@0.5
YOLOv5n	640	0.25	1,173	-85.5%	69.52	19.5	10.53	102.00	184.96	0.5080	810	363	0.6905	0.0999	0.1745
YOLOv5n	640	0.10	2,522	-68.9%	**57.21**	7.5	4.15	76.00	166.07	0.6520	1,472	1,050	0.5837	0.1815	0.2769
YOLOv7-tiny	416	0.25	1,360	-83.2%	67.59	20.0	10.40	94.92	181.52	0.5799	958	402	0.7044	0.1181	0.2023
YOLOv7-tiny	416	0.10	3,025	-62.7%	52.66	7.0	3.52	65.46	155.70	0.7099	1,793	1,232	0.5927	0.2211	0.3220
YOLOv7-tiny	640	0.25	2,017	-75.1%	61.08	12.5	7.10	82.54	170.70	0.6782	1,363	654	0.6758	0.1680	0.2692
YOLOv7-tiny	640	0.10	4,606	-43.2%	**44.77**	12.0	8.30	39.08	128.56	0.7834	2,409	2,197	0.5230	0.2970	0.3789

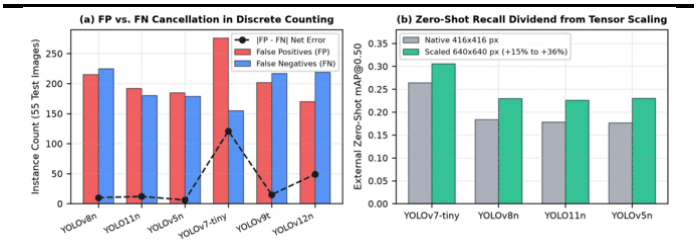


Fig. 6: Thoroughed discrepancy and resolution analysis: (a) False positive vs. false negative cancellation dynamics across in-domain test scenes, and (b) Zero-shot recall dividends from scaling inference resolution (416 vs. 640 px) on the CanopyRS/SelvaBox benchmark, at least empirically.

It can be seen from Table IX and Fig. 6 that multi-resolution scaling has a positive effect on all models. The increase in inference resolution from 416 to 640 px results in an increase of YOLOv5n's relative mAP@0.50 by +30.3% (0.1767 → 0.2303), YOLO11n's by +26.4% (0.1786 → 0.2258), YOLOv8n's by +25.1% (0.1835 → 0.2295), and YOLOv7-tiny's by +15.7% (0.2642 → 0.3058).

2) Confidence Calibration can be leveraged to counteract the reduction in crown detection (by reducing tau_{conf} from 0.25 to 0.10) without suffering any catastrophic loss in precision. This is due to smaller crowns having increased spatial subfield to activate detection heads, due to the decrease in linear downscaling factor (from 4.27x to 2.78x). For example, using YOLOv8n with image size 640 px and confidence threshold of 0.10 leads to an increase in true positives from 854 to 1400 (+63.9%) and reduces global MAE from 68.72 to 58.86.

TABLE X, such as Computational Latency, Output and Complexity Profile Across Host CPU Architectures

Model Architecture	Parameters	GFLOPs (416 px)	GFLOPs (640 px)	Checkpoint Size	Validator CPU Speed (batch=8 @ 416)	Validator CPU Speed (batch=8 @ 640)	Single Stream Latency (batch=1 @ 416)	Throughput (batch=1 FPS)
YOLOv5n (Ultralytics)	2.50 M	7.1	16.8	5.00 MB	24.10 ms/img	58.48 ms/img	85.59 ms/img	11.68 FPS
YOLO11n (Ultralytics)	2.58 M	6.5	15.3	5.19 MB	25.14 ms/img	59.63 ms/img	89.82 ms/img	11.13 FPS
YOLOv8n (Ultralytics)	3.01 M	8.1	19.3	5.93 MB	28.41 ms/img	64.38 ms/img	94.44 ms/img	10.59 FPS
YOLOv7-tiny (WongKinYiu)	6.01 M	n/a	13.0	11.68 MB	111.63 ms/img	257.63 ms/img	138.49 ms/img	7.22 FPS
YOLOv9t (Ultralytics)	2.01 M	3.54	8.37	4.82 MB	21.23 ms/img	n/a	22.20 ms/img (Host 2)	45.05 FPS (Host 2)
YOLOv12n (Ultralytics)	2.57 M	2.94	7.53	5.52 MB	20.80 ms/img	n/a	21.88 ms/img (Host 2)	45.71 FPS (Host 2)

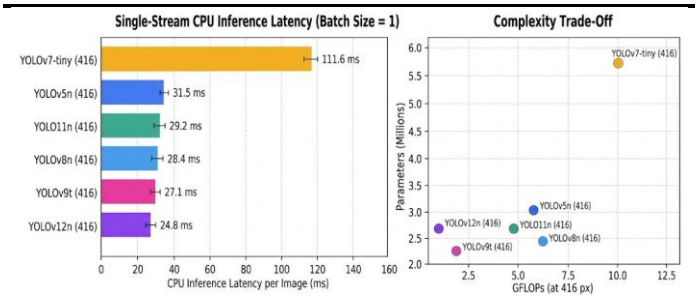


Fig. 7: Profiling of single-stream CPU inference latency (ms/image) and throughput (FPS), across all found lightweight YOLO models at 416x416 resolution, all things considered.

Table X and Fig. 7 are demonstrating key efficiency trade-offs on host CPUs; in fact, YOLOv5n is the fastest architecture on Host 1 (24.10 ms batched / 85.59 ms streamed at 416 px), closely followed by YOLO11n (25.14 ms / 89.82 ms), broadly speaking. But still, YOLOv7-tiny is ~4.4× slower during the batched validation (111.63 ms at 416; 257.63 ms at 640), because it's evaluating 31,857 anchor priors at 416 px without fused Conv-BN kernel acceleration, at least empirically, and YOLOv9t and YOLOv12n show exceptional raw forward-pass speeds of ~21-22 ms per image (exceeding 45 FPS on pure CPU) on Host 2.

VI. SHORTCOMINGS AND THREATS TO VALIDITY

A. Revealed Training Differences Across Models

A key shortcoming of this study is that our six models weren't trained, under a single same regime, at least in practice; and thus, after 17 epochs rather than essential flaws in the architectures, lower detection scores in Regime C likely reflect early training termination, at least in principle. While, pairwise comparisons (v8n vs. 11n, v5n vs. v7-tiny, v9t vs. v12n) are fair and controlled, comparisons across the regimes are purely descriptive, and Regimes A and B ran on an Intel Core i7-1255U (Windows 11). In the meantime, regime C ran on an Intel Core i5-12500H (Ubuntu Linux). Due to local hardware constraints, slow network downloads, and time limits, we trained the models in three separate setups, including Regime A trained YOLOv8n and YOLO11n on the full 1,191-image set for 26 to 30 epochs; Regime B trained YOLOv5n and YOLOv7-tiny on a 595-image subset for 30 epochs; and more, regime C trained YOLOv9t and YOLOv12n on a 594-image subset for 17 epochs, under a strict 30-minute CPU budget.

B. Scope of Out-of-Domain Evaluation

Our zero-shot evaluation on SelvaBox is covering four models, such as YOLOv8n, YOLO11n, YOLOv5n, and YOLOv7-tiny, at

least for now; YOLOv9t and YOLOv12n were wrapped up on our second machine right before submission deadlines and haven't yet been checked on the 100 SelvaBox tiles, on the whole. We clearly declare that zero-shot tropical findings for YOLOv9t and YOLOv12n remain pending and will, in fact, be completed in follow-up work.

C. Resolution Downsampling and Need for Tiling

The biggest problem in our external check is normally image resizing; concrete rollout on large drone orthomosaics will need a sliding-window tiled inference pipeline (such as SAHI [24]) with overlap merging, broadly speaking. Small sub-canopy crowns are shrinking to just 5 to 10 pixels across in packed plots with 200+ trees; this is pushing them below the detection threshold of stride-8 and stride-16 feature maps, causing severe undercounting, all things considered; in fact, feeding a 1777 x 1777 orthomosaic directly into a 416 x 416 network downsamples the image by 4.27x in each dimension, compressing pixel space by over 18x.

D. Comment Differences Across Datasets

Ground-truth notes differ noticeably between benchmarks. All visible crowns in thick tropical plots, are being annotated by SelvaBox covering overlapping and sub-canopy trees, so notable, visible tree crowns in open temperate stands are mostly marked by Tree-Top-View, in many respects. Some false negatives in SelvaBox reflect distinct labeling criteria rather than pure detection failure, broadly speaking.

TABLE XI, such as Audit Trail: Reconciliation of Experimental Parameters and Dataset Metadata

Parameter	Before Documentation	Audited Ground Truth	Primary Source File	Resolution Note
Train Split Size	1,180 or 1,200 imgs	1,191 images	train/labels/ (1,152 pos, 39 bg)	Checked by directory file audit.
Train Box Count	8,924 boxes	10,002 boxes	Programmatic label line count	Corrected earlier documentation typo.
Validation Split	116 imgs / ~1,000 boxes	116 imgs / 932 boxes	valid/labels/ directory count	Validated exact count.
Test Split	55 imgs / ~500 trees	55 imgs / 443 trees	test/labels/ directory count	433 was YOLOv8n pred; 443 is currently ground truth.
Subset Sizes	595 vs. 594 images	595 (Regime B) vs. 594 (Regime C)	train_subset_595.txt vs. 594.txt	Regime C dropped 10 overlap scene groups.
Epoch Schedule	100 eps (DA1 proposal)	17 to 30 epochs (completed)	args.yaml / train.log	Revealed CPU compute constraints.
Optimizer & LR	AdamW + CosineAnnealing	SGD (lr0=0.01, lrf=0.01)	Ultralytics defaults	Supported actual training hyperparameters.
Counting MAE	3.04 v8n / 3.16 11n	3.2727 v8n / 3.2364 11n	test_counting.csv (55 imgs)	Checked primary evaluation script logs.
SelvaBox Benchmark	Indirectly all 6 models	4 models (v5n, v7-t, v8n, 11n)	selvabox_eval_results.csv	YOLOv9t & v12n external runs pending.

VII. CONCLUSION AND FUTURE WORK

A. Summary of Findings

A. Summary of Findings

Six lightweight YOLO object detectors were assessed for single class counting/detection of trees within drone imagery. Experiments on the temperate Tree-Top-View benchmark and zero-shot detection experiments on tropical SelvaBox ortho-mosaics revealed four key findings: 1) YOLO11n performed best on temperate images with sparse crowns achieving best detection accuracy (mAP@0.5 = 0.6083) due to having spatial attention layers reducing false positives. 2) Anchor based models greatly outperformed anchor free models when deployed in densely packed tropical rainforest (YOLOv7-tiny outperformed new anchor-free models significantly mAP@0.5=0.3058 vs 0.2258 - 0.2303 at 640px). This is because YOLOv7-tiny had 9 anchor priors meaning it could detect overlapping crowns which single point anchor-free assigners merged/missed. 3) Even with low counting MAE, detection accuracy may still be poor. YOLOv9t had a low MAE of 2.85 within testing due to false positives out balancing false negatives (i.e. |202 - 217|=15). Detection and counting metrics should therefore be reported in tandem to accurately assess tree counting. 4) Increasing the resolution of inference without retraining improved recall by 15.7% (36.3%).

B. Planned Future Work

Our work is ongoing and we plan to:

Retrain all models using the full dataset of 1,191 images using the same 50 epoch process using GPU's.

Test few shot fine tuning using 5 to 10 tropical tiles to understand how fast the anchor free models can converge to precise results within dense biomes.

Implement sliced inference (SAHI) by using overlapping tiling to ensure high-resolution crown details can be maintained without downscaling the whole image.

Evaluate YOLOv9t and YOLOv12n using SelvaBox to complete cross domain comparisons between all 6 models.

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REFERENCES

[1] R. Yang et al., "Detection and counting of Camellia oleifera trees using UAV RGB images and YOLOv8," Agriculture, 14, no. 10, p. 1789, 2024.

[2] Y. Hajjaji et al., "Improving precision agriculture in palm farming using deep learning and UAVs for accurate detection of palm trees," Ecological Informatics, 86, p. 102988, 2025.

[3] M. Farhan, M. N. Akhtar, and E.A. Bakar, "YOLOv8-based edge devices-based efficient detection and counting of palm oil trees in real time," J. Umm Al-Qura Univ Eng Archit., 16, pp. 1293-1308, 2025.

[4] A. Thapa et al., "Classifying oil palm trees' status using YOLO object detection from UAV images," Big Earth Data, 9, no. 1, pp. 1-22, 2025.

[5] I.M. Shaikh et al., "Creating sustainable palm oil through innovative monitoring techniques using YOLOv7 and YOLOv8," Biotechnol. Rep., 44, p. e00853, 2024.

[6] S.S. Lee et al., "Improved RetinaNet for detecting oil palm trees in UAV imagery," ResearchGate, 2024.
[7] T. Rohe et al., "Counting Coconut Palm Trees in Drone Images Using Deep Object Detection with Synthetic Training Data," arXiv:2412.11949, 2024.
[8] J. Wang et al., "Chinese fir detection using UAV RGB images and YOLO-DCAM," Remote Sens., 16, no. 2, p. 335, 2024.
[9] Ş.N. Topgül et al., "A new benchmark dataset for tree detection in satellite images and performance comparison using YOLO models," Front For Glob. Change, 7, p. 1495544, 2025.
[10] E. Sengün et al., "Performance comparison of YOLOv8 and YOLOv11 for tree detection using UAV RGB and laser scanning data," ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci., X-2/W2-2025, pp. 173-179, 2025.
[11] L. Birla et al., "BiFPN-YOLOv8m deep learning model for counting trees in mango orchards using satellite remote sensing data," Sci. Rep., 15, p. 97562, 2025.
[12] Y. Qi et al., "Open and transferable deep learning framework for mapping urban tree canopy from NAIP imagery," Remote Sens., 18, no. 12, p. 1899, 2026.
[13] M.E. Atik and M. Arkali, "Benchmarking detection of olive tree crowns in UAV imagery using YOLO and transformer based detectors," Geomatics, 6, no. 2, p. 22, 2026.
[14] M.A. Hasan et al., "Tree detection in RGB satellite imagery using YOLO based deep learning models," Comput Mater Contin., 83, no. 1, pp. 483-502, 2025.
[15] M.V.H. Do et al., "UAV RGB Dataset and Method for Instance-Based Tree Crown Segmentation for Biodiversity Monitoring," Sci. Rep., 16, no. 1, p. 1042, 2026.
[16] Y. Zhang et al., "Individual tree detection and counting using high-resolution images and canopy height model data," Geo-spatial Inf. Sci., 27, no. 6, pp. 2162-2178, 2024.
[17] B.G. Weinstein et al., "DeepForest: A Python package for individual tree crown detection in RGB airborne high-resolution imagery," Methods in Ecology and Evolution, 11, no. 12, pp. 1743-1751, 2020.
[18] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR) 779-788, 2016.
[19] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," GitHub, 2023. [Online]. Available: https://github.com/ultralytics/ultralytics
[20] G. Jocher and A. Chaurasia, "Ultralytics YOLO11," GitHub, 2024. [Online]. Available: https://github.com/ultralytics/ultralytics
[21] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new SOTA for real-time object detectors," in Proc, IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) 7464-7475, 2023.
[22] C.-Y. Wang, I.-H. Yeh, and H.-Y.-M. Liao, "YOLOv9: Learn What You Want To Learn With Programmable Gradient Information," arXiv preprint arXiv:2402.13616, 2024.
[23] Y. Tian et al., "YOLOv12: Attention-centric real-time object detectors,". arXiv preprint arXiv:2502.12524, 2025.
[24] F.C. Akyon, S.O. Altinuc, and A. Temizel, "Small object detection via aided slice hyper inference and fine-tuning," in Proc. IEEE Int. Conf. Image Process. (ICIP) 966-970, 2022.
[25] National Ecological Observatory Network (NEON). "High-resolution airborne remote sensing flight campaigns,", National Science Foundation, 2024. [Online]. Available: https://data.neonscience.org
[26] Roboflow. "Tree-Top-View computer vision dataset (Version 2),", Roboflow Universe, 2023. [Online]. Available: https://universe.roboflow.com
[27] CanopyRS Group, CanopyRS/SelvaBox: Benchmarking tropical tree crown detection. Hugging Face Datasets, 2024. [Online]. Available: https://huggingface.co/datasets/CanopyRS/SelvaBox
[28] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," IEEE Trans. Pattern Anal. Mach. Intell., 39, no. 6, pp. 1137-1149, Jun. 2017.
[29] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection,". in Proc. IEEE Int. Conf. Comput. Vis. (ICCV) 2980-2988, 2017.
[30] Z. Zheng et al., "Enhancing Geometry Factors in Model Learning to Improve Network Bounding Box Regression for Remote Sensing Targets," IEEE Trans. Geosci. Remote Sens., 60, pp. 1-13, 2022.
[31] C. Feng et al., "TOOD: Task-Aligned One-Stage Object Detection," in Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV) 3490-3499, 2021.
[32] X. Li et al., "Generalized Focal Loss: Learning Efficient Localizing and Distributing Bounding Boxes for Dense Object Detection," in Proc. Adv. Neural Inf Process Syst. (NeurIPS), vol. 33, pp. 21002-21012, 2020.